Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH. CE and VIIth
Name of the Paper: Ground Improvement Technique
Paper Code: TCE-722

Time: 1.5 hour

Maximum Marks: 50

Note:

(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.
(iii) Please specify COs against each question.

Q1. (10 Marks)

a. What are the most effective ground improvement techniques for stabilizing structures built on black cotton soils. (CO1)

OR

b. What are the key roles of ground improvement techniques in ensuring the stability of foundation structures, and how do they contribute to reducing settlement issues? (CO1)

Q2. (10 Marks)

a. What is the principle behind in situ densification methods like vibration at the ground surface for granular soils? (CO2)

OR

b. Compare vibro-floation and dynamic compaction Methods. (CO 2)

Q3. (10 Marks)

a. Explain briefly with a neat sketch the vibro compaction method of densifying deep granular deposits. (CO2)

OR

b. Describe the use of vertical drains, such as sand drains and sand wick geo-drains, in accelerating consolidation in cohesive soils. (CO2)

Q4. (10 Marks)

a. Compare the Chemical and Mechanical Stabilization of Soil. (CO1)

OR

b. Discuss the methods of Improving for soil by Excavating and Replacing. (CO1)

Q5. (10 Marks)

a. When selecting ground improvement techniques for a construction project, how do the varying properties of alluvial, laterite, and black cotton soils influence the choice of method. (CO1)

OR

b. How do thermal methods work in the densification of cohesive soils, and what types of soil conditions make them necessary? (CO2)